

MM-RAG Reference Document

This is a synthetic reference document used for evaluating the MM-RAG retrieval pipeline. It contains both plain text and embedded raster figures, so retrieval quality can be measured for both text-only and figure-required questions.

The manufacturing team tracks quarterly widget output across four product lines, labeled A through D. Line B produced the most units this quarter, while Line C lagged behind the others. Figure 1 shows the full breakdown of quarterly widget output.

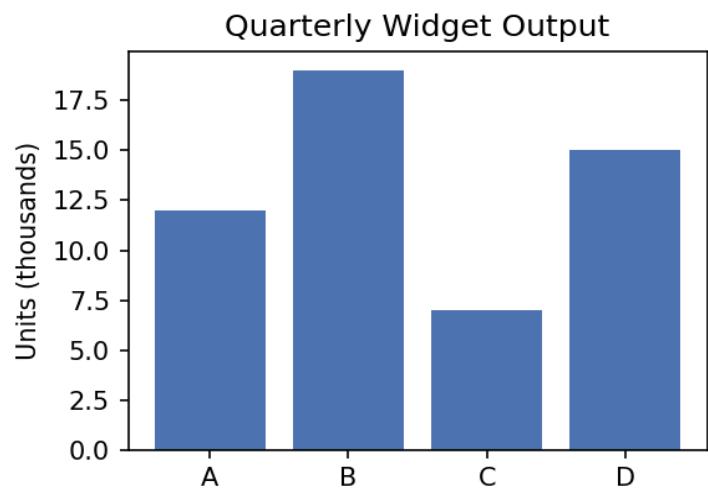


Figure 1: Quarterly widget output by product line.

System Architecture

The retrieval pipeline processes an uploaded document in six stages: ingestion, indexing, retrieval, fusion, reranking, and generation. Each stage is implemented as an importable Python function so it can run inline during a live upload, rather than as a batch script tied to a fixed folder on disk.

Figure 2 below shows this pipeline as a sequence of stages. Fusion combines the text and image ranked lists using Reciprocal Rank Fusion with a constant k of 60. Reranking then applies a cross-encoder to the text candidates only, since no off-the-shelf cross-encoder exists for text-image pairs at this scope.

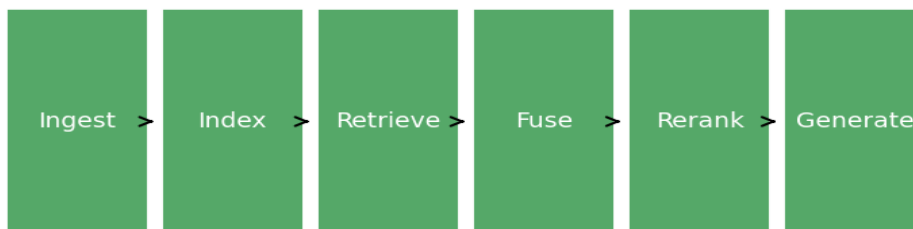


Figure 2: The six-stage MM-RAG pipeline.